Ayush Sharma

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Education

Indian Institute of Technology Goa 2023 – Present
B.Tech. in Mathematics and Computing *Goa, India*

Experience

Software Development Intern May 2025 – June 2026
Cogniway *Remote*

- Designed and implemented an automated therapy session scheduler that accounts for therapist availability, leave schedules, qualification matching, workload balancing, conflict-free allocation, and session-spacing constraints.
- Built REST APIs and MongoDB models, and developed an AI-assisted goal management module using Gemini AI for hierarchical therapy planning, customizable assessment scales, therapist approval workflows, and patient progress tracking.
- **Tech Stack:** React, Node.js, Express.js, MongoDB

Projects

Fault-Tolerant Task Processing System | *Node.js, Express.js, PostgreSQL, Redis* | GitHub Dec. 2025 – June 2026

- Built a distributed asynchronous task queue designed for reliable delivery, fault tolerance, and duplicate prevention using Redis-backed queue management and PostgreSQL.
- Implemented multi-layer idempotency, exponential-backoff retries, and zombie-task recovery to improve reliability of asynchronous processing.
- Benchmarked approximately 2K tasks/sec enqueue throughput with 20 ms average API latency (53 ms p99); tested exactly-once execution across 1.5K+ tasks with four concurrent workers.
- Eliminated a TOCTOU race by replacing a check-then-insert workflow with an atomic database operation.

Cache Simulator | *C++, STL, Cache Design* | GitHub Sept. 2025 – Nov. 2025

- Built a configurable, two-level set-associative cache simulator to model how cache design decisions affect memory performance across workload patterns.
- Implemented O(1) LRU replacement, write-back/write-through policies, write-allocate/no-write-allocate policies, and next-line prefetching using STL data structures.
- Reduced cache misses by up to 50% on sequential traces using next-line prefetching; validated correctness with eight manually derived test cases covering AMAT, hit rate, and writebacks.

Secure Endpoint Monitoring Platform | *Go, PostgreSQL, React, nftables, Linux* May 2026 – Present

- Initiated an OS-level exam-integrity system for supervised Linux labs to monitor activity that browser-based proctoring may not capture.
- Developing monitoring for terminal commands and Vim activity, network restrictions to the exam server, and evidence collection through a PostgreSQL backend; an HMAC-chained log helps detect tampering.
- Built a React dashboard for reviewing flagged activity. In VM-lab tests, Vim-monitoring bypass attempts were flagged within approximately 10 seconds.
- Load-tested 20 simulated clients at approximately 245 events/sec, achieving 0.27-second batch p99 latency.

Technical Skills

Languages: C++, Python, JavaScript, Go, SQL
Frameworks & Technologies: React.js, Node.js, Express.js, REST APIs
Databases & Messaging: PostgreSQL, Redis, MongoDB
Tools & Platforms: Git, Postman, Vercel, Render, Linux, nftables
Technical Concepts: Data Structures and Algorithms, OOP, Operating Systems, DBMS, Computer Architecture, Computer Networks, Memory Hierarchy

Achievements & Extracurricular

- Achieved a LeetCode rating of 1900+ (top 5% globally).
- Secured a top-32 position among 100+ teams at the Goa Police Hackathon 2024 with a fraud detection system reported to achieve 85%+ accuracy.
- Represented IIT Goa at the Inter-IIT Basketball Championship.